INCH-POUND

MIL-R-1688F
26 April 1990
SUPERSEDING
MIL-R-1688E
24 May 1989

MILITARY SPECIFICATION

ROPE, FIBROUS, CLIMBING TYPE, NYLON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers nylon rope used for mountaineering operations.

1.2 Classification. The rope shall be furnished in the following sizes as specified (see 6.2):

Sizes (Circumference)

3/4 inch
1-1/8 inches
1-1/4 inches
1-1/2 inches

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4020

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:37Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

UU-T-81 - Tags, Shipping and Stock

MILITARY

MIL-C-3131 - Cordage; Packaging Of

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:37Z
Check the source to verify that this is the current version before use.

3.2 Government and contractor purchases. The requirements specified in 3.9 and 3.10 apply only to rope purchased directly by the Government. All other requirements apply both to rope purchased by a contractor as a component for an end item and to rope purchased directly by the Government.

3.3 Standard sample. The rope shall match the standard for shade and appearance and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.4 Materials. The rope shall be fabricated from a minimum of 6 denier per filament, bright, virgin, continuous filament, light- and heat-resistant polyamide (nylon) having a minimum 6.5 grams per denier strength. The nylon shall be a long chain polymer made of hexamethylene diamine and adipic acid or its derivatives, or a long chain polymer of epsilon amino caprioic acid. Mixtures of nylon fiber types (6 or 66) in any one rope shall be prohibited. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.5 Construction and physical requirements. The rope shall be of three-strand construction and conform to the requirements specified herein and in table I. Each strand shall be made of the same size of three ply yarns. All strands shall have an equal numbers of yarns. The singles yarn shall be between 2500 and 8000 denier. The singles yarn, when measured in the three ply, shall be left-hand or "S" lay and shall have a minimum of 22 turns per foot. The lay of the rope shall be right-hand or "Z" lay. The rope shall be heat set and shall not show more than a 5 percent residual shrinkage when tested as specified in 4.4.4. The rope shall be fabricated without knots or splices in the singles yarns, ply yarns, strands, or rope.

TABLE I. Physical requirements

Sizes (circumference (inches)	Tolerance, (inches)	Diameter approximate 1/ (inches)	Length, feet per lb, minimum 2/	Breaking strength, minimum (pounds)	Elongation, minimum (percent)	Length per 10 turns	
						maximum (inches)	minimum (inches)
3/4	+ 1/8 - 1/16	1/4	47.0	1300	35.0	7.8	7.0
1-1/8	+ 1/8 - 1/16	3/8	23.0	3150	35.0	12.4	11.1
1-1/4	+ 1/8 - 1/16	7/16	17.0	4500	35.0	14.3	12.9
1-1/2	+ 1/8 - 1/16	1/2	8.5	6500	35.0	17.2	15.1

Source: https://assist.dla.mil -- Downloaded: 202640-16T13:37Z
Check the source to verify that this is the current version before use.

1/ Not a specification requirement, included for information only.

2/ A minus tolerance of 10 percent is allowed for dyed rope.

3.5.1 Breaking strength after heat aging. The loss in breaking strength after heat aging of the rope shall not exceed 10.0 percent of the initial breaking strength when tested as specified in 4.4.4.

3.5.2 Hardness. The finished rope shall have a hardness of 20 ± 5 pounds when tested as specified in 4.4.4.

3.6 Extraneous material. No extraneous material shall be added for the purpose of weighting the rope. The extractable matter of the finished rope shall not exceed 4.0 percent when tested as specified in 4.4.4.

3.7 Color. Unless otherwise specified (see 6.2), the rope shall be dyed Olive Drab 7 (see 6.4).

3.7.1 Colorfastness. The dyed and finished rope shall show fastness to weathering equal to or better than the standard sample or equal to or better than a rating of "fair" when tested as specified in 4.4.4.

3.7.2 Matching. The color of the dyed and finished rope shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 ± 200K, with illumination of 100 ± 20 foot candles and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200K.

3.8 pH. The pH value of the rope shall be within the range of 5.0 to 9.0 when tested as specified in 4.4.4.

3.9 Identification. Each coil of rope shall have a ticket (identification tag) attached for identification purposes. The ticket shall conform to the requirements for type B, class 1, size 4 or 5 of UU-T-81. The ticket shall be made of not less than 15-point paper stock and have a minimum tearing resistance in both directions (total) of 850 grams when tested as specified in UU-T-81. All ticket entries shall be legibly printed, stamped, or typed with water insoluble ink. The ticket shall contain the following information:

National Stock number
Nomenclature
Specification number
Length
Size (circumference)
Contract number and date
Contractor's name
Date of manufacture (month and year)

Source: https://assist.dla.mil -- Downloaded: 202610-16T13:37Z
Check the source to verify that this is the current version before use.

3.10 Put-up. Unless otherwise specified (see 6.2), the rope shall be put-up in coils in lengths specified in table II for the respective sizes (circumference). The rope, when put-up in coils as specified in table II, shall be in one continuous piece and so wound that each turn and layer is free from entanglement. The ends of all rope shall be cut off squarely and securely whipped or heat sealed to prevent fraying or untwisting.

TABLE II. Put-up coil length

Rope size (circumference in inches)	Coil length (feet)	
	minimum	maximum
3/4	196	204
1-1/8	196	204
1-1/4	117	123
1-1/2	117	123

3.11 Workmanship. The end item shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:37Z
Check the source to verify that this is the current version before use.

4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3 and tested for the characteristics specified in 4.4.4.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Component and material certification. A certificate of compliance may be acceptable as evidence that the characteristics listed in table III conform to the specified requirements.

TABLE III. Component tests

Characteristic	Requirement paragraph
Material	3.4
Denier per filament	3.4
Grams per denier strength	3.4
Light and heat resistant yarn	3.4
Fiber types mixtures	3.4
Strand construction	3.5
Denier of singles	3.5
Lay of singles	3.5
Turns per foot (singles)	3.5

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:37Z
Check the source to verify that this is the current version before use.

4.4.2 End item visual examination. The rope shall be examined for the defects listed in table IV. All defects shall be counted regardless of their proximity to each other. The lot size shall be expressed in units of coils of rope. The sample unit shall be one coil of rope. One hundred feet shall be examined on each coil of rope. The inspection level shall be II (see 6.5), and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.5 for major defects and 4.0 for total (major and minor combined) defects. Any critical defect found during sampling shall be cause for rejection of the lot represented by the sample.

TABLE IV. End item visual defects

Examine	Defect	Classification		
		Critical	Major	Minor
Appearance and workmanship	Any cut	1		
	Chafed or damaged	2		
	Kink	3		
	Knot (singles, ply, strand, or rope)	5		
	Broken or loose end (singles, ply, strand, or rope)	6		
	Bulged strand	7		
	Spliced or otherwise secured to make a continuous length	8		
Color	Other than specified		101	
	Resists and/or spotty dyeing			201
	Specific shade not within established tolerances			202
	Uneven			203
Cleanness	Spot or stain, clearly visible 1/			204
Identification	Omitted, incorrect, illegible, or not as specified			205

1/ At normal inspection distance (approximately 3 feet).

4.4.3 End item length and winding examination. The rope shall be examined for the defects listed below. The lot size shall be expressed in units of coils of rope. The sample unit shall be one coil of rope. Defects shall be as listed below. The inspection level shall be S-3, and the AQL, expressed in terms of defects per hundred units, shall be 4.0. The required length of the coil (see table II) shall be determined using the following formula:

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:37Z
Check the source to verify that this is the current version before use.

W x L = Length per coil in feet.
When W = the net weight of the coil in 0.1 pound.
L = the lot average of the weight-length relationship (length/pound) as specified in Table I.

Defects

Length of rope is less than specified or more than length specified (see table II).
Length of rope on coil is more than 6 feet less than the length indicated on the ticket.
Improperly or not firmly wound resulting in kinking, knotting, entangling, or slippage during unwinding, or otherwise affecting free unhampered unwinding of rope.
Any end frayed or not cut off squarely or neatly.
Any end not securely whipped or heat sealed.

4.4.4 End item testing. The end items shall be tested for the characteristics listed in table V. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table V shall be followed. When the data in the "Number of determinations" and "Results reported as" columns are not specified in table V, they shall be as required by the referenced test method. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The lot size shall be expressed in units of coils of rope. The sample unit shall be one coil of rope. The sample size shall be in accordance with table VI. The lot shall be unacceptable if one or more sample units fail to meet any requirement. All test reports shall contain the individual values utilized in expressing the final results. Tests to determine compliance with document requirements may be made under prevailing atmospheric conditions. In case of dispute, tests shall be made upon material that has reached equilibrium under standard conditions as defined in FED-STD-191.

TABLE V. End item tests

Characteristic	Requirement reference	Test method	Number of determinations per sample unit	Results reported as
Number of strands	3.5	Visual	1	Pass or fail
Lay of rope	3.5	Visual	1	Pass or fail
Shrinkage	3.5	6010	-	-

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:37Z
Check the source to verify that this is the current version before use.

TABLE V. End item tests (cont'd)

Characteristic	Requirement reference	Test method	Number of determinations per sample unit	Results reported as
Circumference	Table I	6003	-	-
Length per pound	Table I	6004	-	-
Breaking strength:				
Initial	Table I	6015 1/	-	-
After heat aging (percent strength)	3.5.1	6015 and 4.5.1	2	Average of 2 determinations to nearest 10 lb. calculated to nearest 0.1 percent
Elongation	Table I	6015	-	-
Length per 10 turns	Table I	6000	-	-
Hardness	3.5.2	6020	-	-
Extractable matter	3.6	2611 2/	2	Average of 2 determinations to nearest 0.1 percent
Colorfastness to weathering	3.7.1	5671 3/	1	Pass or fail
pH	3.8	2811	-	-

1/ Failures in tests tend to occur at or near the end of the free length, but may occur in the mid-span between the splices. Failures occurring within the eye splice shall be disregarded and another specimen shall be tested from the same sample unit. (See figure 1.)

2/ Except that a 5-gram specimen shall be used and the extract solvent shall be petroleum ether.

3/ The specimens shall be exposed for 40 standard fading hours.

Source: https://assist.dla.mil -- Downloaded: 202610-16T13:37Z
Check the source to verify that this is the current version before use.

TABLE VI. Sample size

Lot size (coils of rope)	Sample size (sample units)
15 and under	2
16 up to and including 40	3
41 up to and including 110	5
111 up to and including 300	7
301 up to and including 500	10
501 and over	15

4.4.5 Packaging inspection. The sampling and inspection of the preservation, packaging, and container marking shall be in accordance with the requirements of MIL-C-3131, except that the inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.5 Methods of inspection.

4.5.1 Breaking strength after heat aging. Two breaking strength specimens, prepared as specified in Method 6015 of FED-STD-191, shall be heated for 5 days in a convection air oven at 175° ± 2°F. After aging, the specimens shall be allowed to reach equilibrium under standard conditions before being tested for strength. Breaking strength shall be determined in accordance with Method 6015 of FED-STD-191. The percent change in strength shall be calculated based on the average strengths initially and after heat aging.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A and Commercial. The rope, put-up as specified in 3.10, shall be preserved in accordance with the applicable requirements of MIL-C-3131.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Levels A, B, and Commercial. The rope shall be packed in accordance with the applicable requirements of MIL-C-3131.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:37Z
Check the source to verify that this is the current version before use.

5.3 Marking. In addition to any special markings required by the contract or purchase order, interior packages and shipping containers shall be marked in accordance with MIL-C-3131.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The rope is intended to be used for mountaineering operations.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Size (circumference) required (see 1.2).
c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1).
d. When a first article is required (see 3.1, 4.3, and 6.3).
e. Color, if other than Olive Drab 7 (see 3.7).
f. Put-up, if other than specified (see 3.10).
g. Levels of preservation and packing (see 5.1 and 5.2).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Standard sample. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Subject term (key word) listing.

Coils
Cordage
Equipment, mountaineering

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:37Z
Check the source to verify that this is the current version before use.

Custodians:

Army - GL
Navy - MC

Review activities:

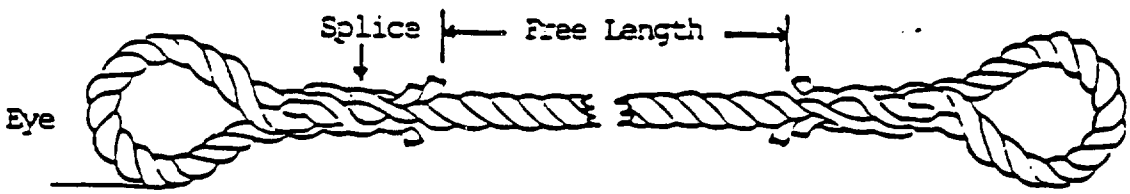
Army - AR, MD
Navy - AS
DLA - IS

Preparing activity:

Army - GL

(Project 4020-0319)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:37Z
Check the source to verify that this is the current version before use.



Eye Splice On 3-Strand Twisted Cordage

FIGURE 1. Schematic of typical rope failure positions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:37Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-R-1688F	90/04/26

3. DOCUMENT TITLE
ROPE, FIBROUS, CLIMBING TYPE, NYLON

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. Army Natick RD&E Center	(1) Commercial 508-651-5221	(2) AUTOVON 256-5221
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-ES Natick, MA 01760-5014	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 2025-10-16T13:37Z
Check the source to verify that this is the current version before use.